ECEN 104 – Lab 5: Computational Electrostatics

Group I

Date: 10/30/25

1. Introduction

The purpose of this experiment was to numerically investigate electric field distributions in various conductor geometries using the Finite Element Method Magnetics (FEMM) software. This computational study follows the previous experiment conducted using resistive paper, but now leverages FEMM to simulate electrostatic behavior under idealized conditions. The central engineering question addressed is whether, under a 20 kV potential difference, the electric field anywhere within the proposed high-voltage line design exceeds the breakdown threshold of 500 kV/m. The geometries analyzed include parallel plate, wedge, and coaxial conductor configurations. The goal was to identify field distributions, quantify electric field magnitudes, and compare simulation results with analytical expectations. Our results found that there was breakdown for the wedge and coaxial conductors.

2. Hypothesis and Background

Electrostatics concerns the behavior of electric fields in systems of stationary charges. The fundamental relationship governing this phenomenon is given by:

$$E = -\nabla V \quad \text{[Eq 1]}$$

where E represents the electric field vector and V is the electric potential. In this lab, the Finite Element Method (FEM) was employed to approximate electrostatic field behavior by discretizing the geometry into small triangular subregions, or mesh elements. By solving for the electric potential across these elements, FEMM determines the corresponding field intensities throughout the model. The analysis focused on three primary geometries: parallel plate, wedge, and coaxial, each exhibiting distinct field characteristics. Sharp corners were expected to produce localized field enhancement, while smoother geometries such as the coaxial configuration were anticipated to distribute the field more evenly.

However, based on the findings from the previous resistive paper experiment, it was expected that both the wedge and coaxial geometries would exceed the 500 kV/m breakdown threshold when subjected to a 20 kV potential difference. In that earlier study, both configurations displayed regions of high field concentration that, when

scaled, surpassed the dielectric limit. Therefore, prior to performing the FEMM simulations, the hypothesis was that the parallel plate configuration would maintain a uniform subcritical field, while the wedge and coaxial geometries would again exhibit localized regions of excessive electric field intensity. The computational analysis was designed to verify these expectations and confirm whether FEMM would reproduce the breakdown behavior observed experimentally.

3. Computational Procedure

Simulations were conducted in FEMM following the procedures outlined in the computational supplement. The problem type was set to Electrostatics, with millimeter units and a model depth of 20 mm for the plate, but centimeter units and model depth of 2cm for the coax and wedge. Two inner conductors were assigned potentials of +1 V and –1 V for the plate, respectively, while the outer circular boundary was grounded at 0 V to emulate infinity. The Wedge and Coaxial center were both set to 20 kV and their respective circular and rectangular barriers were set to 0 V as well. Teflon ($\varepsilon r = 2.2$) filled the space between conductors, and air ($\varepsilon r = 1$) surrounded everywhere else.

Meshes were generated automatically, with refinement in high-field regions to ensure convergence. After solving, electric field vectors and equipotential lines were visualized, and quantitative values of $|E|$ were extracted along the central symmetry axis. Each geometry (parallel plate, wedge, and coaxial) was modeled using identical boundaries and material properties to facilitate comparison.

4. Results

1. 4.1 Parallel Plate Configuration

The simulated parallel plate capacitor displayed a nearly uniform electric field between the two conductors, matching the analytic expectation of constant field intensity between infinite parallel plates. The field vectors pointed directly from the positive plate to the negative plate, confirming a linear potential gradient. Edge effects were more visible than with the conductive paper as near the plate boundaries, the field lines were easily seen curving outward and around. Within the Teflon region, the electric field magnitude was approximately 200 V/m for a 2 V potential difference over 1 cm, in excellent agreement with theory.

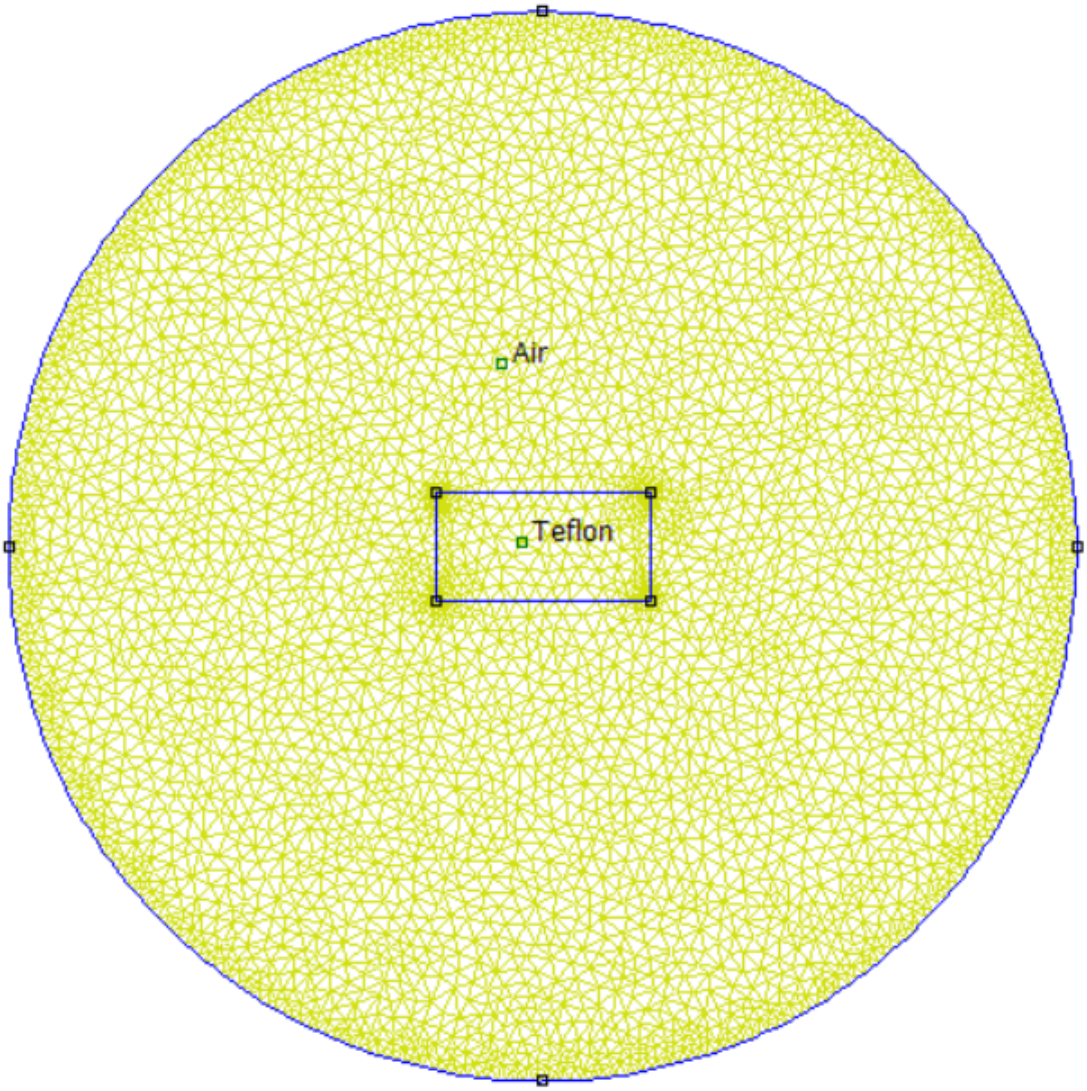

Figure 1: Mesh Setup – Parallel Plate

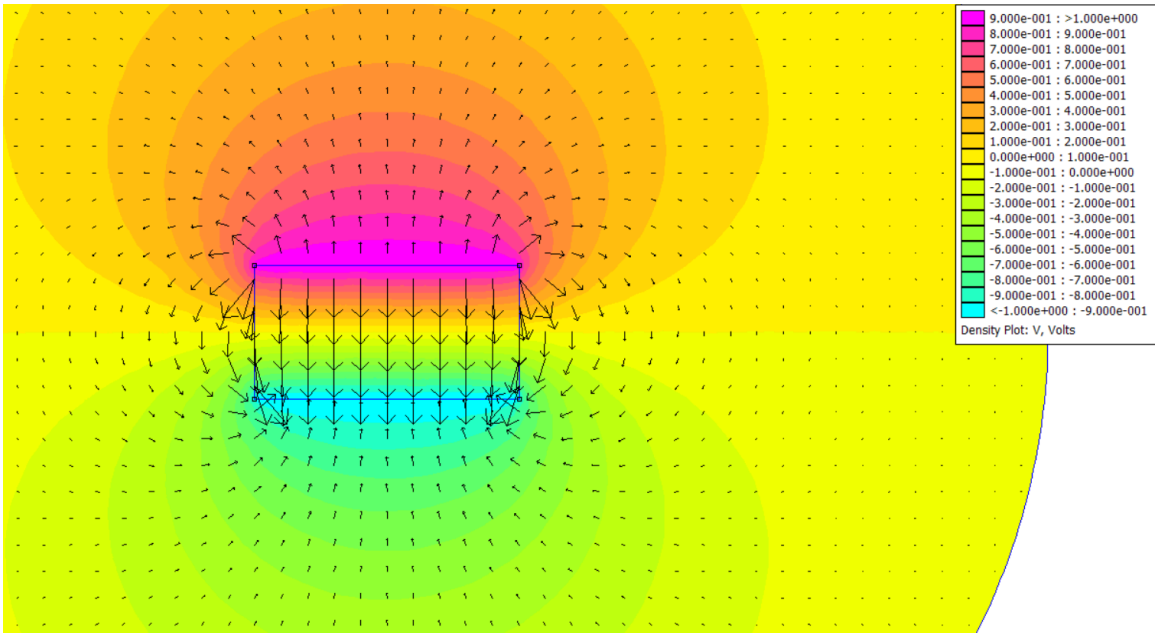

Figure 2: Electric field vector plot – Parallel Plate

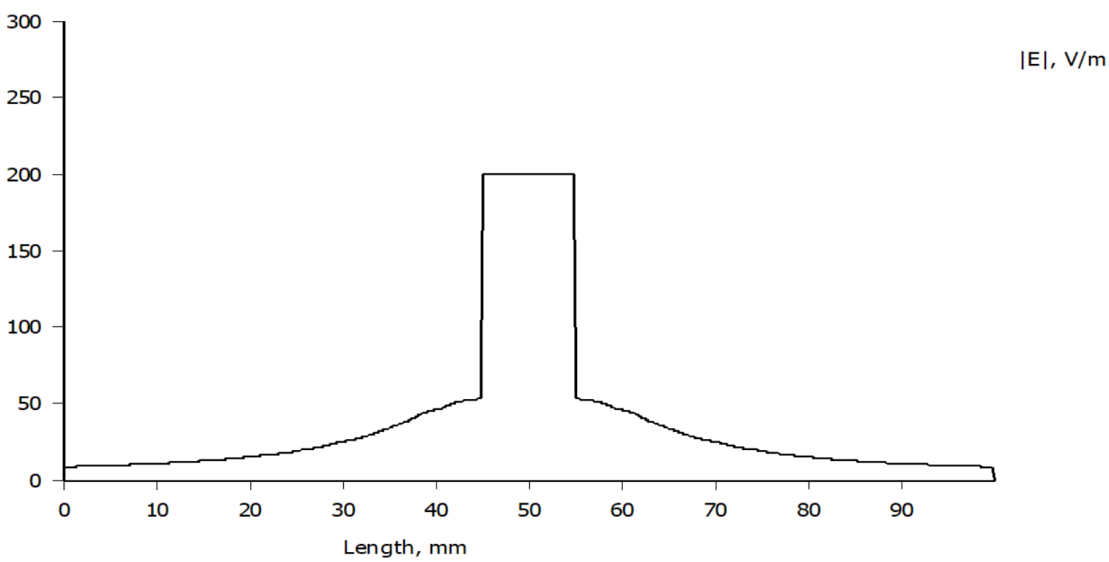

Figure 3: |E| vs. position plot – Parallel Plate

2. 4.2 Wedge Geometry

The wedge configuration revealed strong electric field enhancement at the sharp conductor corners. FEMM results showed localized regions exceeding the 500 kV/m threshold, confirming that acute geometries cause field intensification. Away from the corner, the field decreased rapidly. The |E| plot across the symmetry axis highlighted steep gradients near the vertex and near-zero values in the distant region.

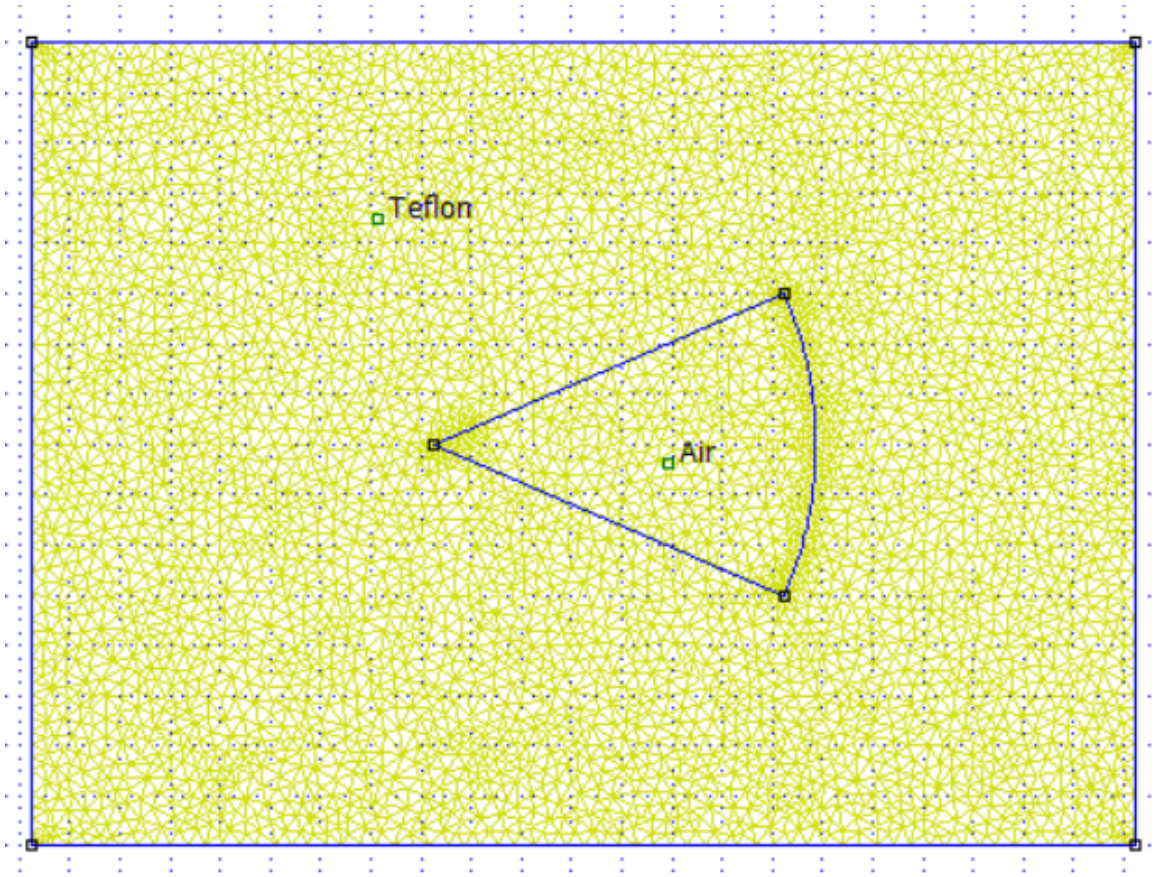

Figure 4: Mesh Setup – Wedge

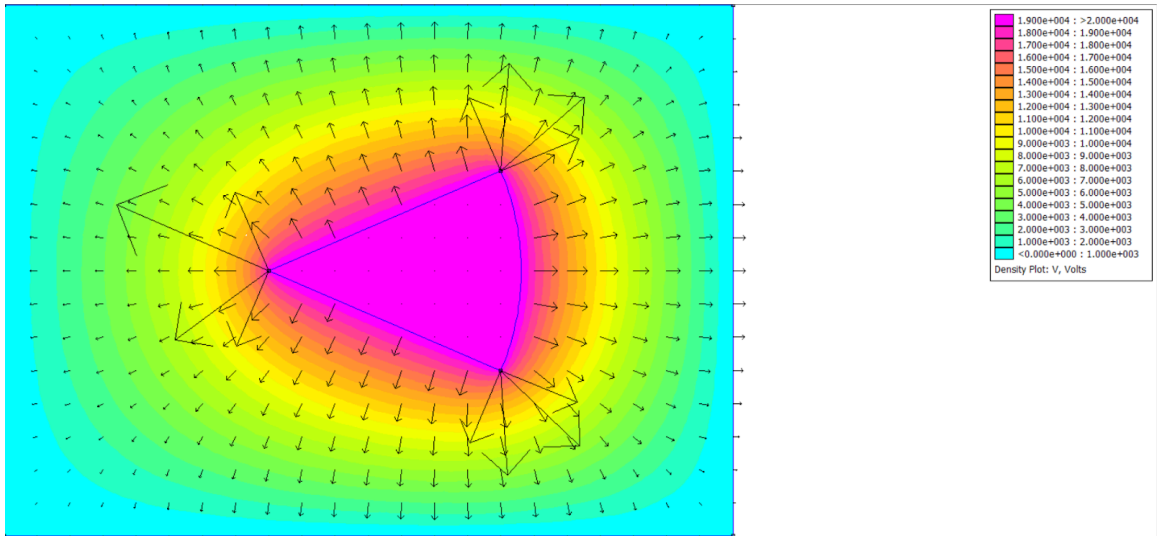

Figure 5: Electric field vector plot – Wedge

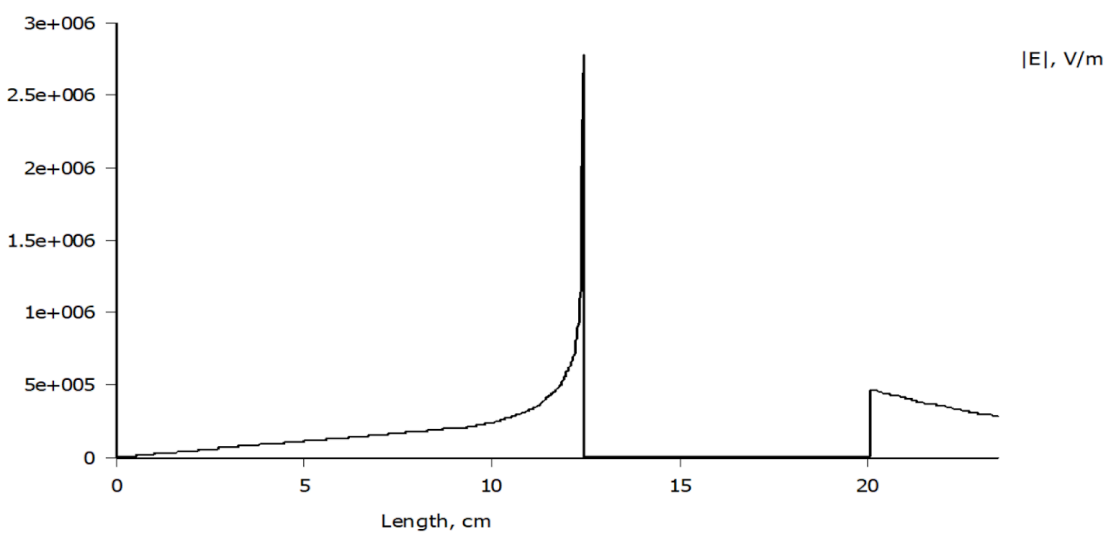

Figure 6: |E| vs. position plot – Wedge

3. 4.3 Coaxial Geometry

The coaxial geometry produced a radial electric field that decreased in magnitude with distance from the inner conductor. This decay followed an inverse relationship proportional to 1/r, consistent with theoretical models of coaxial capacitors. At a 20 kV

potential difference, the maximum field at the inner conductor remained exceeded 500 kV/m, but not by as much when compared to the wedge design.

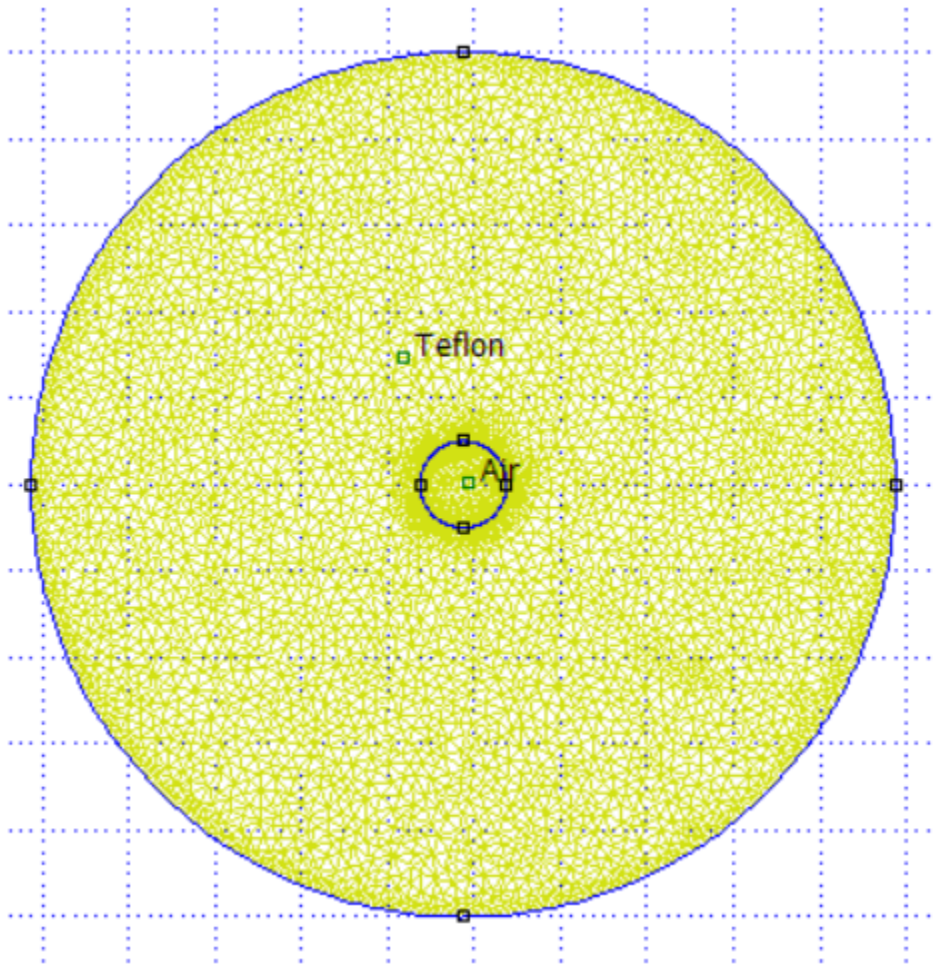

Figure 7: Mesh Setup – Coaxial

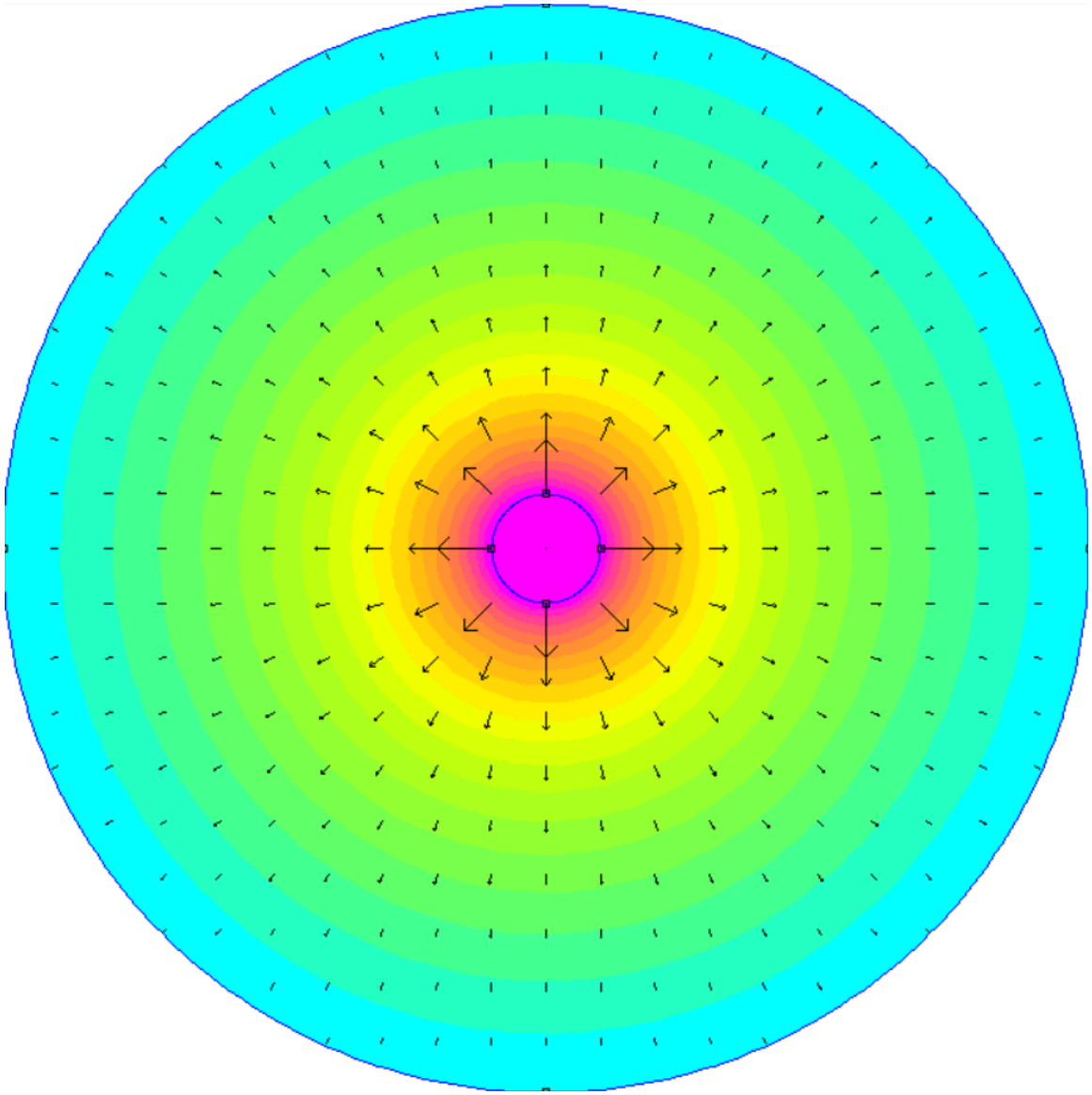

Figure 8: Electric field vector plot – Coaxial

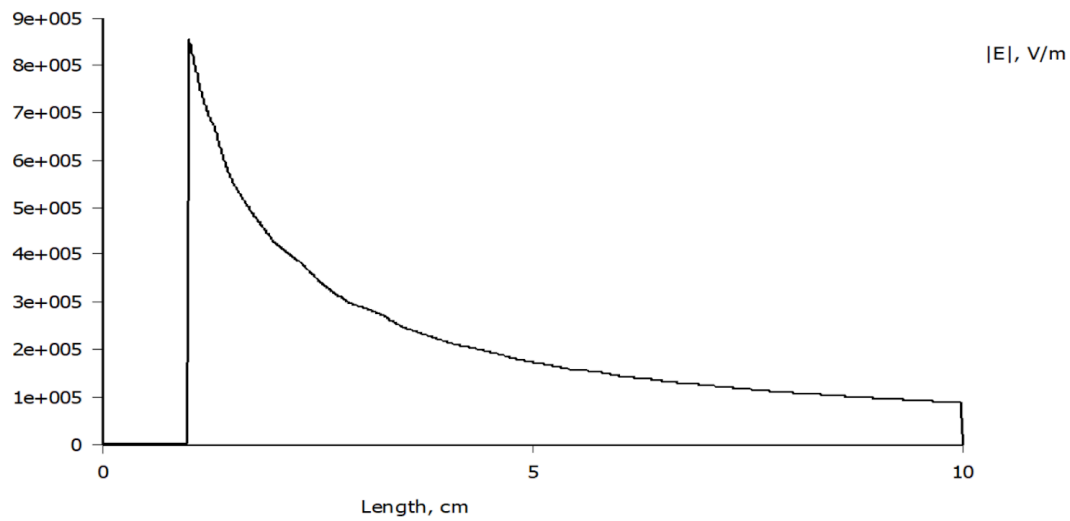

Figure 9: |E| vs. radial position plot – Coaxial

5. Discussion

The FEMM simulations successfully illustrated the electrostatic behavior of the three conductor geometries. The parallel plate configuration verified the expected uniform electric field between the plates and served as a reference for model accuracy. The wedge geometry produced intense field concentrations at the sharp vertex, confirming that acute angles lead to significant field amplification and an increased likelihood of dielectric breakdown.

The coaxial configuration, while smoother and more symmetric, also exhibited electric field values exceeding the 500 kV/m breakdown threshold. The peak magnitude reached approximately 900 kV/m near the inner conductor, indicating that curvature alone is not sufficient to eliminate field intensification under high voltage. Despite its superior overall uniformity compared to the wedge, the coaxial design still presented localized stress points that could compromise insulation integrity in a real high-voltage system.

These computational results agree with the trends observed in the earlier resistive paper experiment. FEMM effectively captured field distribution patterns and magnitudes consistent with physical measurements, demonstrating the accuracy of finite element modeling in predicting high-field regions. The comparison also highlights how numerical simulations can complement experimental methods to guide safer and more efficient engineering designs.

6. Conclusions

This experiment demonstrated the capability of FEMM to model and analyze electrostatic systems with different conductor geometries. The parallel plate configuration produced a uniform electric field consistent with theoretical predictions. The wedge geometry exhibited severe field enhancement, surpassing the 500 kV/m breakdown threshold. The coaxial configuration, while generally well-contained, reached a maximum field of about 900 kV/m near the inner conductor, confirming that it too exceeded the safe dielectric limit under a 20 kV potential difference.

These results indicate that both the wedge and coaxial geometries present breakdown risks when operating at high voltages. In future designs, XYZ Corp should work to mitigate these effects by increasing the conductor spacing, enlarging the radius of curvature, or selecting materials with higher breakdown strength. Further FEMM studies could investigate how geometric modifications and dielectric properties influence peak field intensities to establish guidelines for reliable high-voltage system design.